

Course on Probabilistic Design of Coastal Structures Statistical Modelling of Environmental Conditions **CLIMATE INDEXES (AWT)**

Jared Ortiz-Angulo Cantos

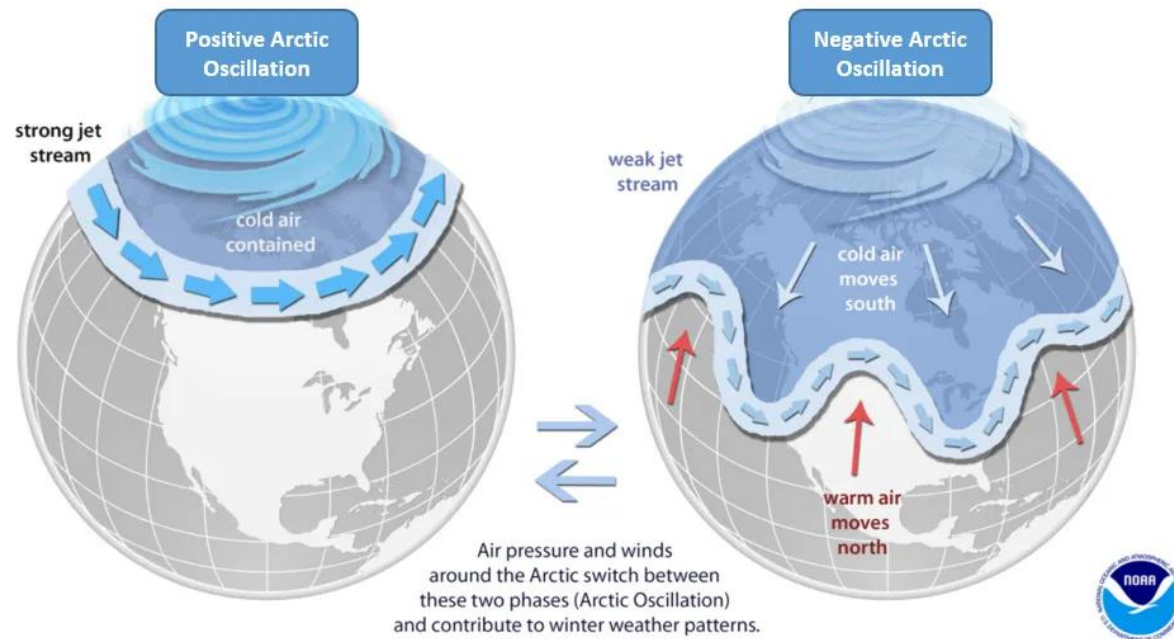
Researcher, GeoOcean Group, Universidad de Cantabria (UC)

ProCoast 2025
October 20th-23th, 2025

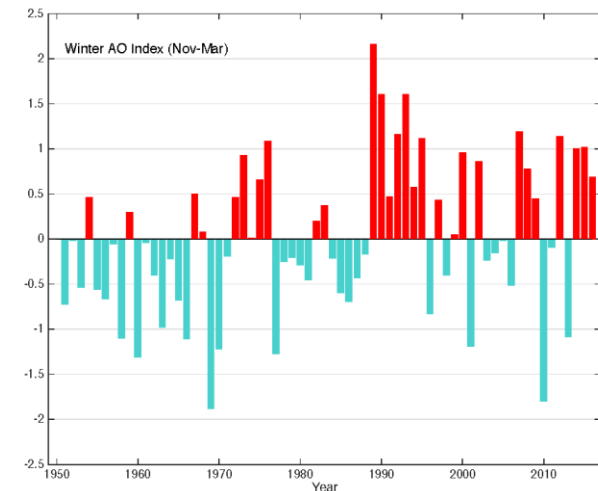
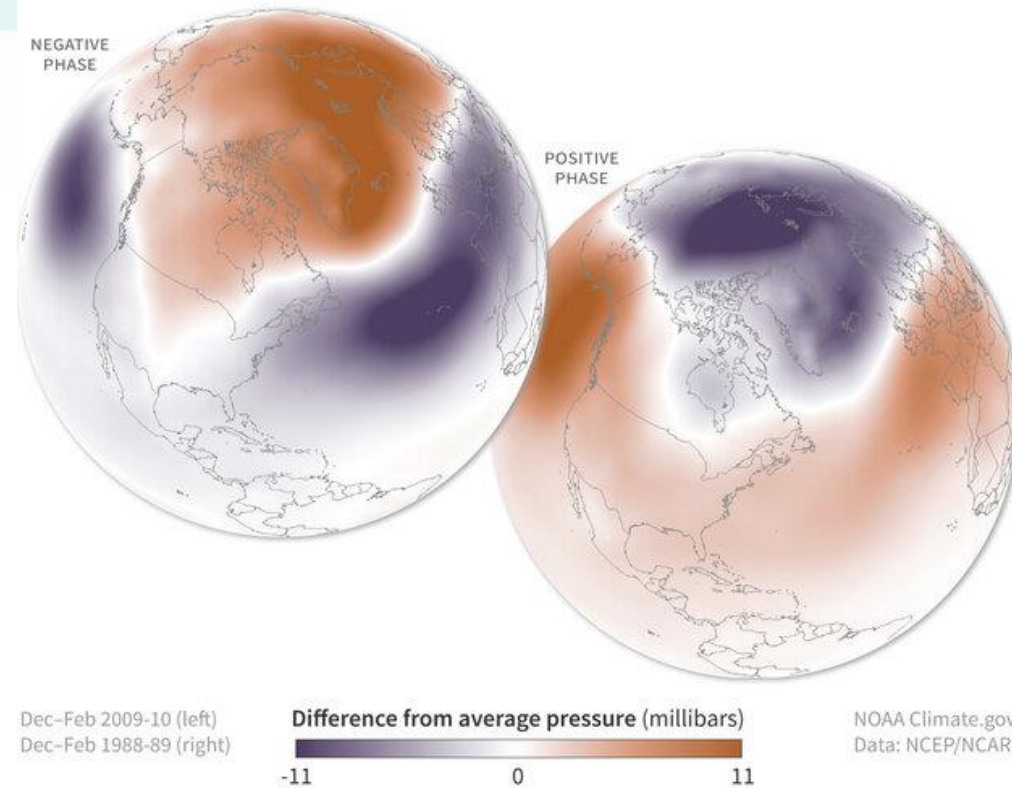


AO: Arctic Oscillation

The AO is a climate pattern **based on pressure patterns** and characterized by winds circulating counterclockwise around the Arctic. When the AO is in its **positive** phase, a ring of strong winds circulating around the North Pole acts to confine colder air across polar regions. This belt of winds becomes weaker and more distorted in the **negative** phase of the AO, which allows an easier southward penetration of colder, arctic airmasses and increased storminess into the mid-latitudes.



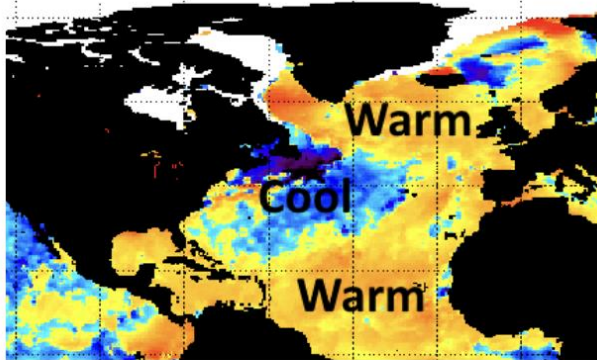
AO PRESSURE PATTERNS



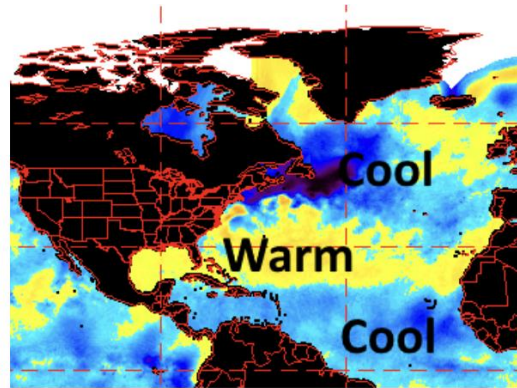
AMO: Atlantic Multidecadal Oscillation

The Atlantic Multi-decadal Oscillation (AMO) is based on the **average anomalies of sea surface temperatures (SST)** in the North Atlantic basin.

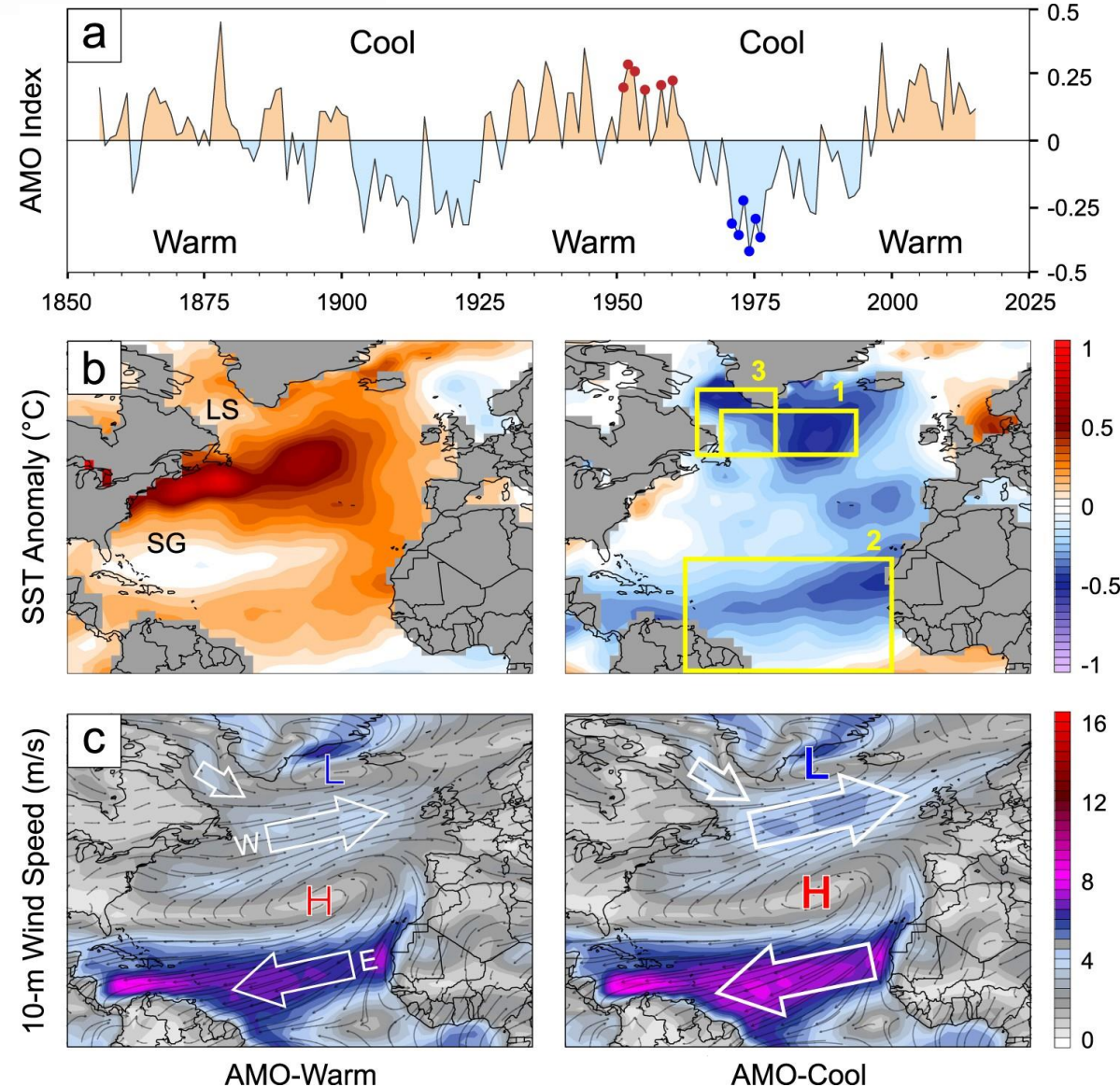
Based on the typical duration of **negative** and **positive** phases of the AMO, the current warm regime is expected to persist at least until 2015 and possibly as late as 2035. Enfield et al. assume a peak around 2020.



An example of a classic positive AMO pattern. Atlantic sea surface temperature anomaly map for June 3, 2010. (Source: NOAA)



An example of a negative AMO pattern. Atlantic sea surface temperature anomaly map for June 1985. (Source: NOAA)



SCA: SCAndinavian pattern

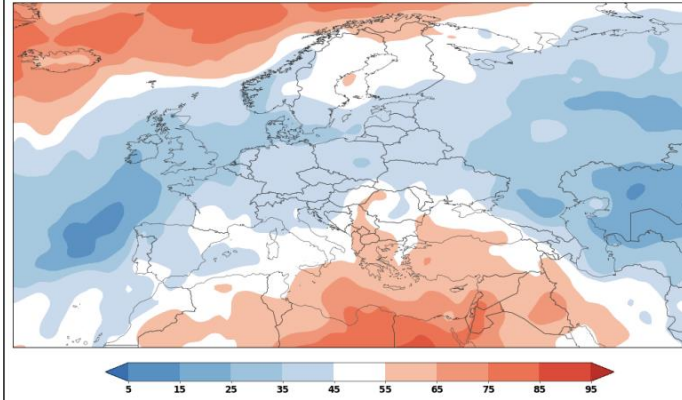
The SCA is an atmospheric climate index (climate cycle or teleconnection) **based on pressure patterns**, that describes flow patterns over Europe.

The Scandinavian index shapes the weather across Europe with characteristic patterns in evidence at different times of the year, making it a valuable input for long-range forecasters such as energy meteorologists.

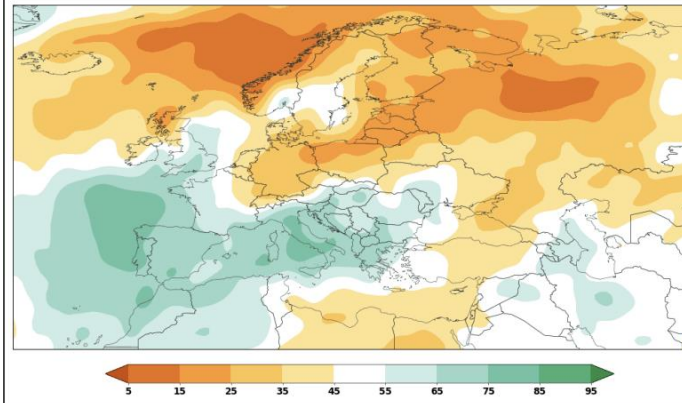
Positive Phase - Winter

Low pressure in Southern Europe and high pressure in Northern Europe

Percent of Years Having Above-Trend DEC-FEB 2m Temperature
1950 1953 1956 1959 1964 1968 1971 1976 1978 1984
1990 1995 2000 2002 2005 2008 2009 2012 2013 2020



Percent of Years Having Above-Normal DEC-FEB Precipitation (ERA5 Reanalysis)
1950 1953 1956 1959 1964 1968 1971 1976 1978 1984
1990 1995 2000 2002 2005 2008 2009 2012 2013 2020



Less Heat in Northern Europe

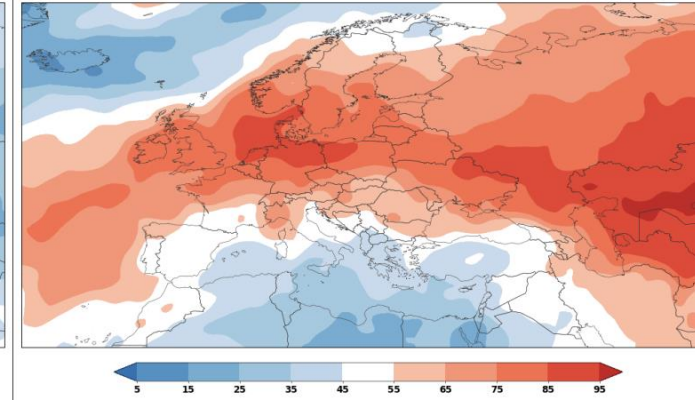
Less Rain in Northern Europe

More Rain in Southern & Western Europe

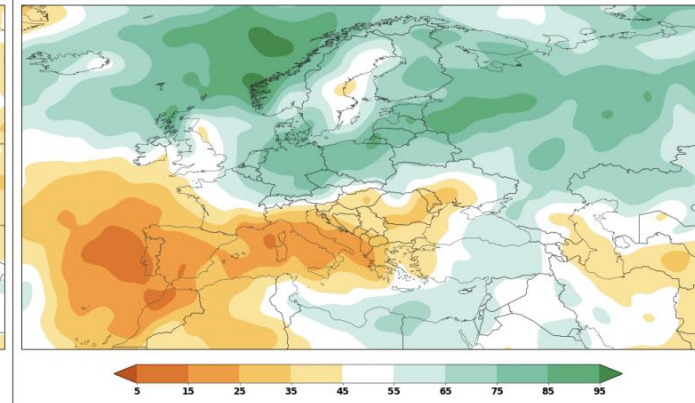
Negative Phase - Winter

High pressure in Central/Southern Europe and low pressure in Northern Europe

Percent of Years Having Above-Trend DEC-FEB 2m Temperature
1961 1967 1974 1975 1980 1982 1988 1991 1992 1994
1998 1999 2001 2003 2006 2014 2015 2019 2021 2023



Percent of Years Having Above-Normal DEC-FEB Precipitation (ERA5 Reanalysis)
1961 1967 1974 1975 1980 1982 1988 1991 1992 1994
1998 1999 2001 2003 2006 2014 2015 2019 2021 2023



More Heat in Northern Europe

More Rain in Northern & Eastern Europe

Less Rain in Western & Southern Europe

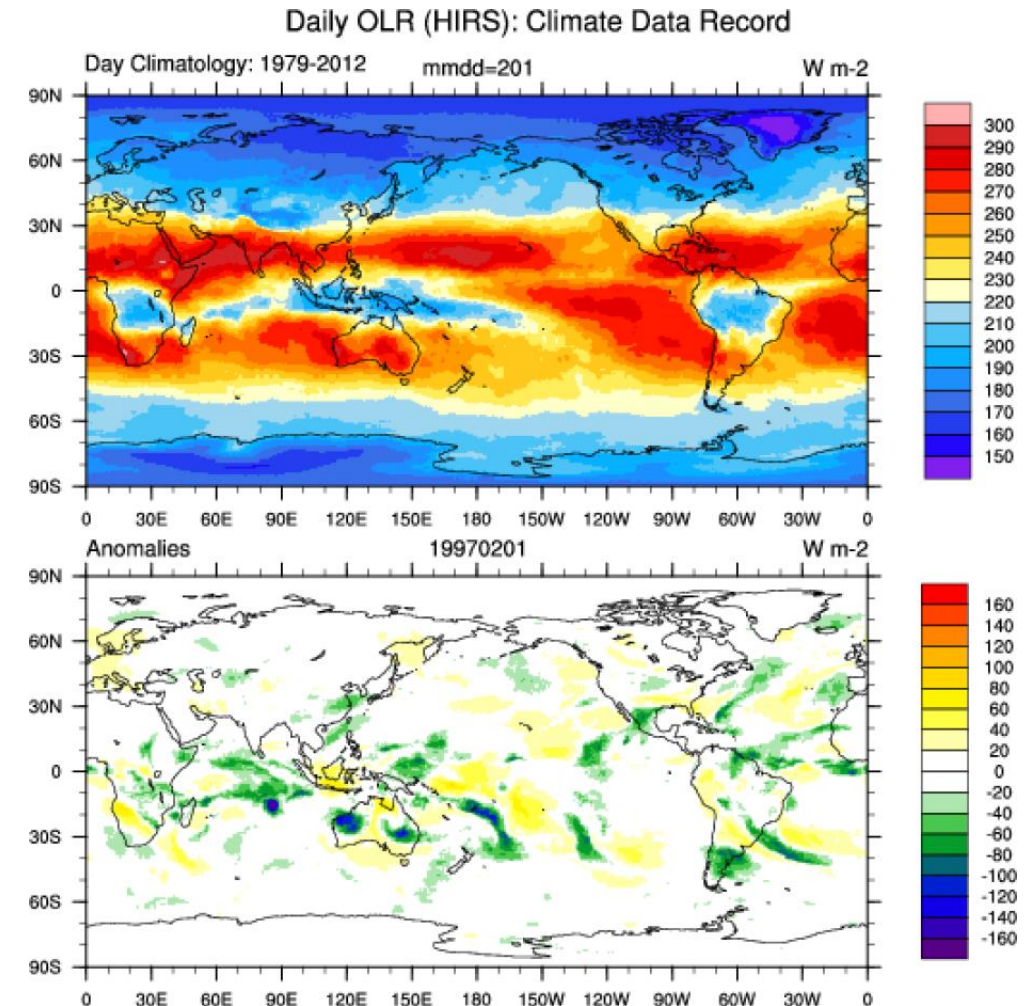
The SCAND is only one factor in a complex system that influences the European climate. Long-range forecasts provided in the WCS Monthly Reports are the best guide to the season ahead.

OLR: Outgoing Longwave Radiation

OLR is a measure of the amount of energy emitted to space by earth's surface, oceans and atmosphere. It's a critical component of the earth's radiation budget.

In a different context, OLR values are often used as a proxy for convection in tropical and subtropical regions since cloud top temperatures (colder is higher) are an indicator of cloud height.

A key component for the standard Madden-Julian Oscillation (MJO) and for climate model evaluation.



PDO: Pacific Decadal Oscillation

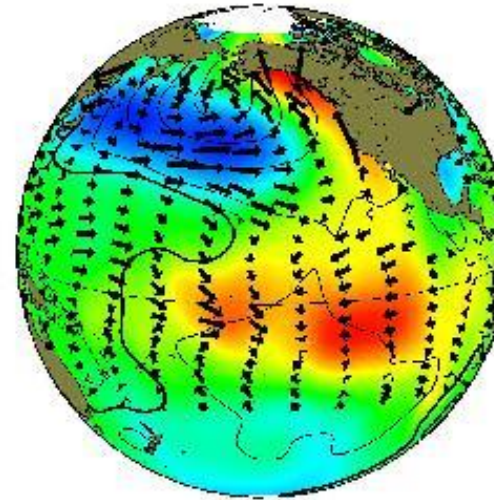
The PDO is based on the **sea surface temperature** in the Pacific Ocean.

The PDO is a robust pattern of ocean-atmosphere climate variability centered over the mid-latitude Pacific basin.

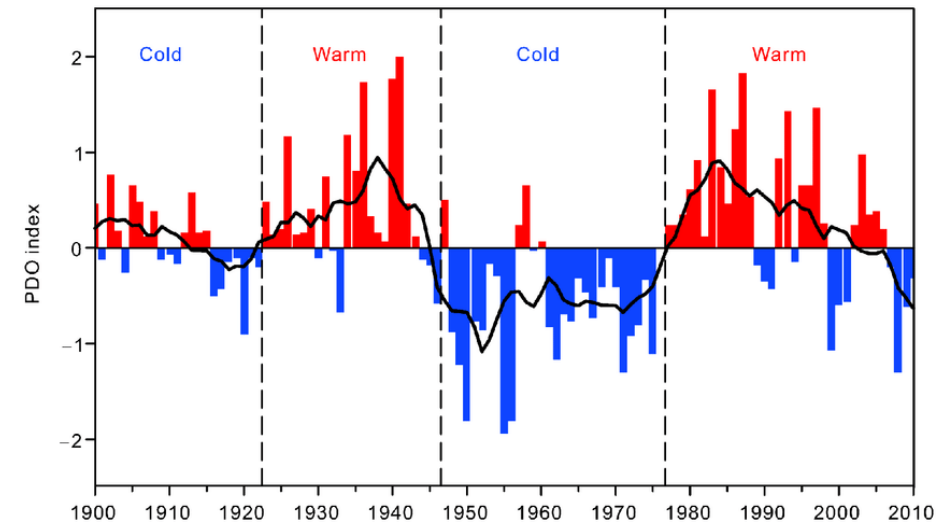
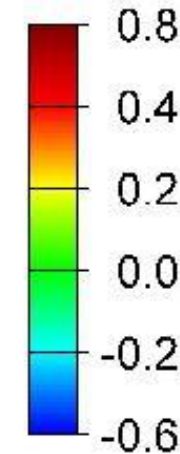
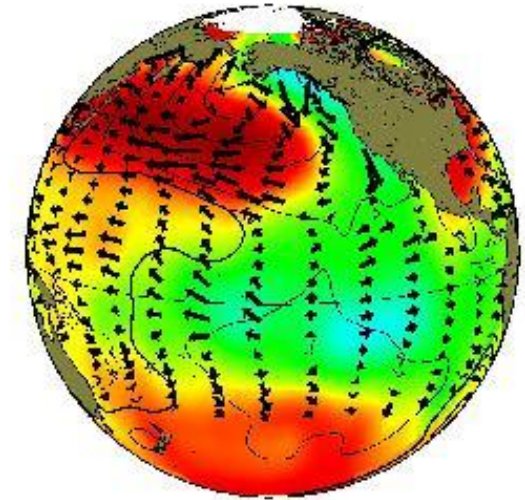
During a "warm", or "positive", phase, the west Pacific becomes colder and part of the eastern ocean warms; during a "cold", or "negative", phase, the opposite pattern occurs.

ENSO teleconnections, the atmospheric bridge

Positive (Warm)



Negative (Cold)



SOI: Southern Oscillation Index

SOI is a standardized index based on the observed **sea level pressure (SLP)** differences between Tahiti and Darwin, Australia.

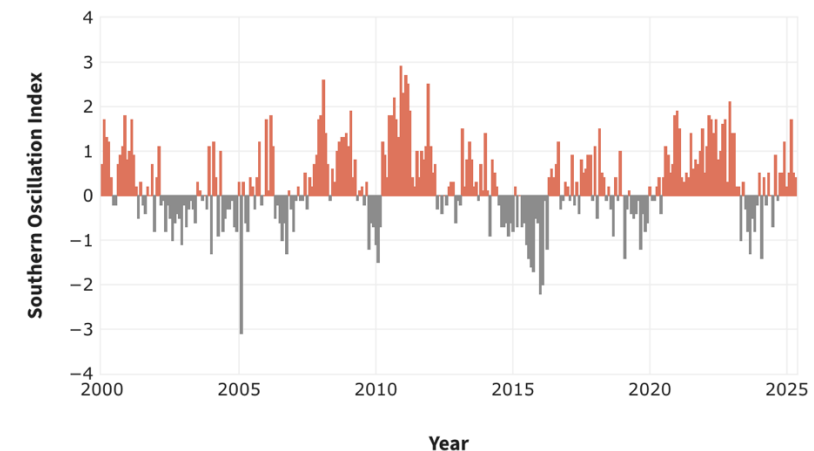
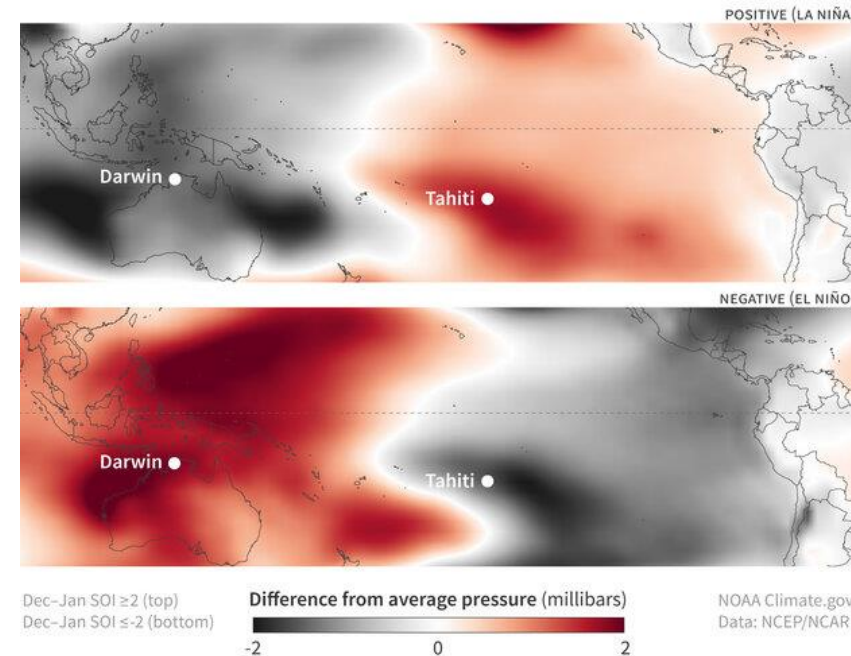
The SOI is one measure of the large-scale fluctuations in air pressure occurring between the western and eastern tropical Pacific (i.e., the state of the Southern Oscillation) during El Niño and La Niña episodes.

The **negative** phase of the SOI represents below-normal air pressure at Tahiti and above-normal air pressure at Darwin. Prolonged periods of negative SOI values coincide with abnormally warm ocean waters across the eastern tropical Pacific typical of El Niño episodes.

Negative = warm ocean waters → El Niño

Positive = cold ocean waters → La Niña

SOI PRESSURE PATTERNS

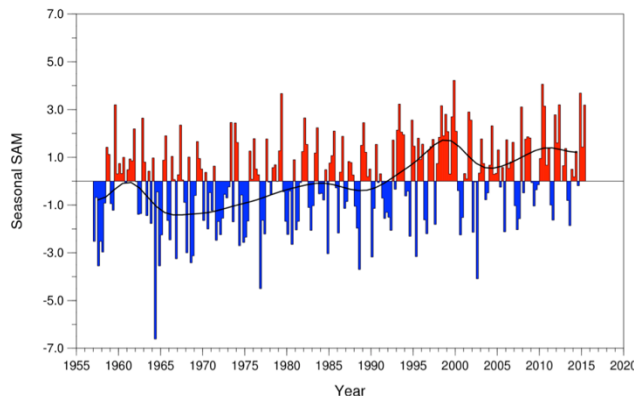


SAM: Southern Annular Mode

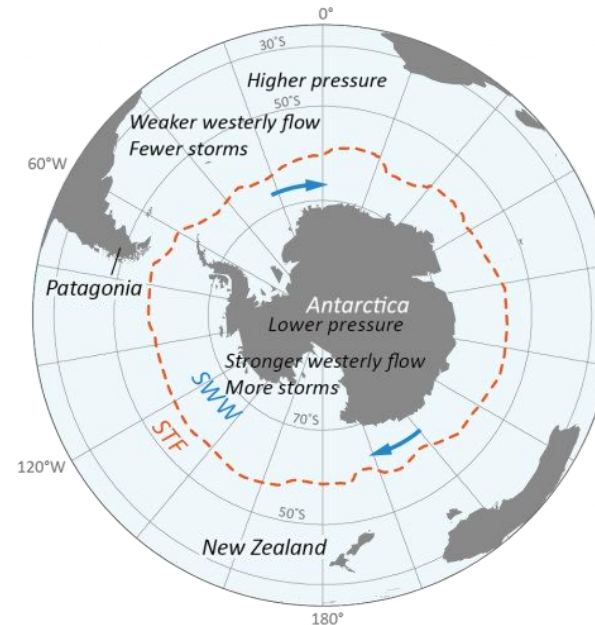
This mode describes low frequency variability of the Southern Hemisphere extratropics.

A **positive** index (lower polar pressure) is associated with weaker, zonal winds in the extratropics.

A **negative** value is associated with stronger zonal winds.



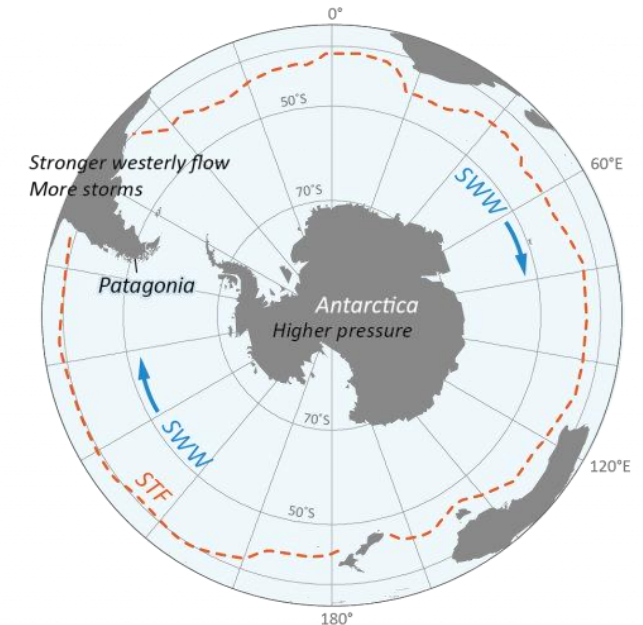
POSITIVE SOUTHERN ANNULAR MODE



SWW and STF contract; increased wind strength

- Winds contract poleward
- Lower anomalous air pressure over Antarctica
- Patagonia warms; decreased precipitation; glaciers recede. Antarctic Peninsula warms.
- Increased CDW upwelling around West Antarctica and Antarctic Peninsula

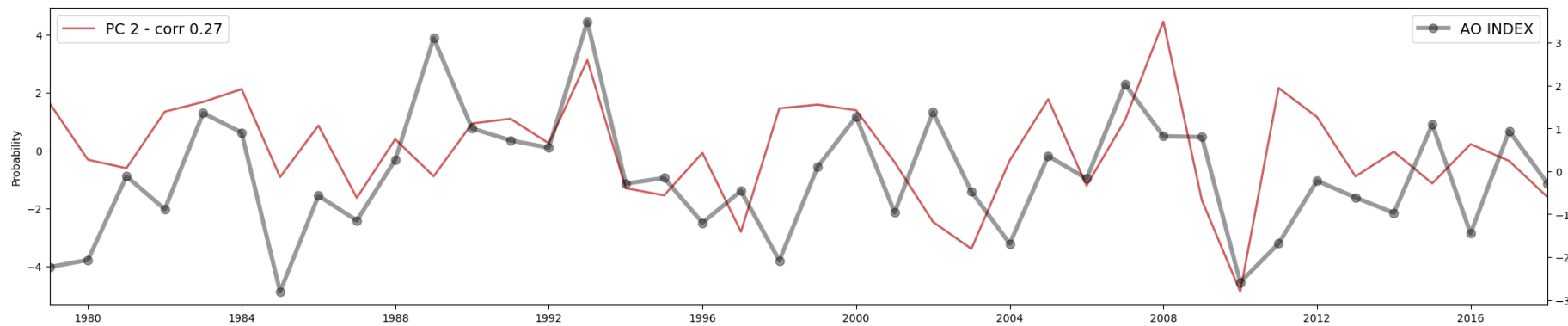
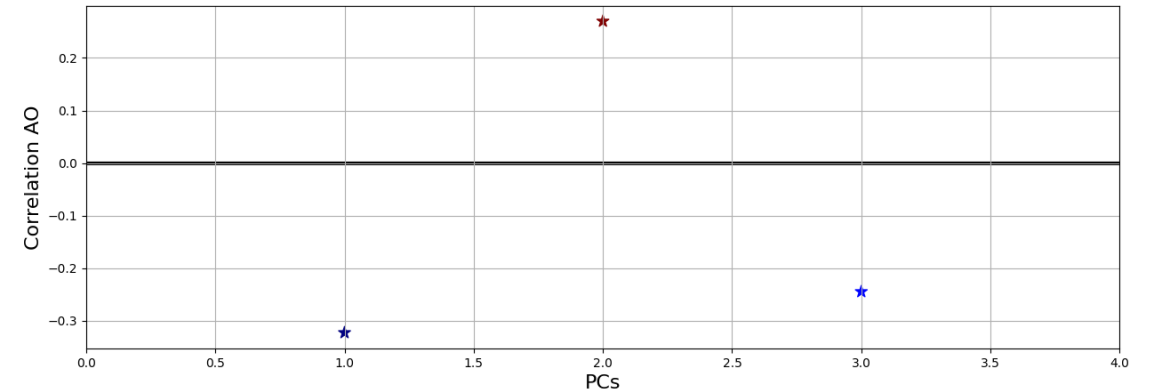
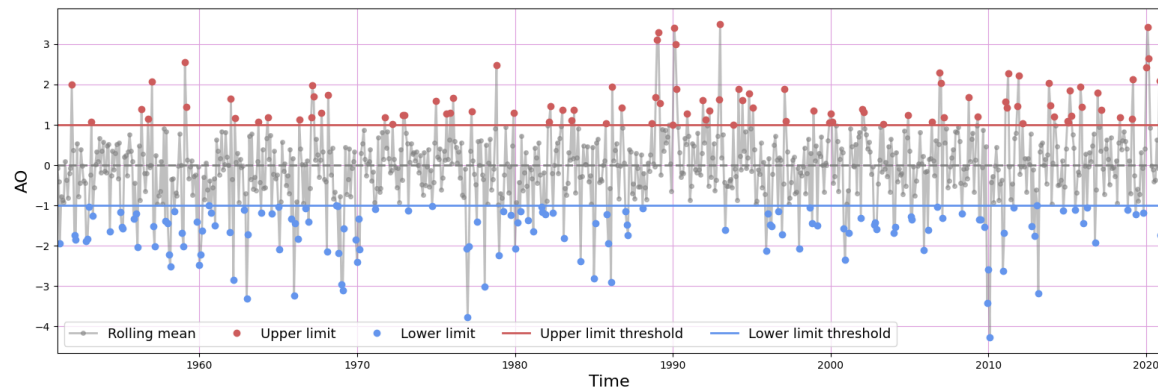
NEGATIVE SOUTHERN ANNULAR MODE



SWW and STF expand; SWW decreased strength

- higher anomalous air pressure over Antarctica
- Patagonia cools; increased precipitation; glaciers advance. Antarctic Peninsula cools.
- Decreased CDW upwelling in West Antarctica and Antarctic Peninsula

Analyze the correlation between the AWTs PCs and the Climate Indexes





Add:

- **NAO**
- **PNA**
- **N34**

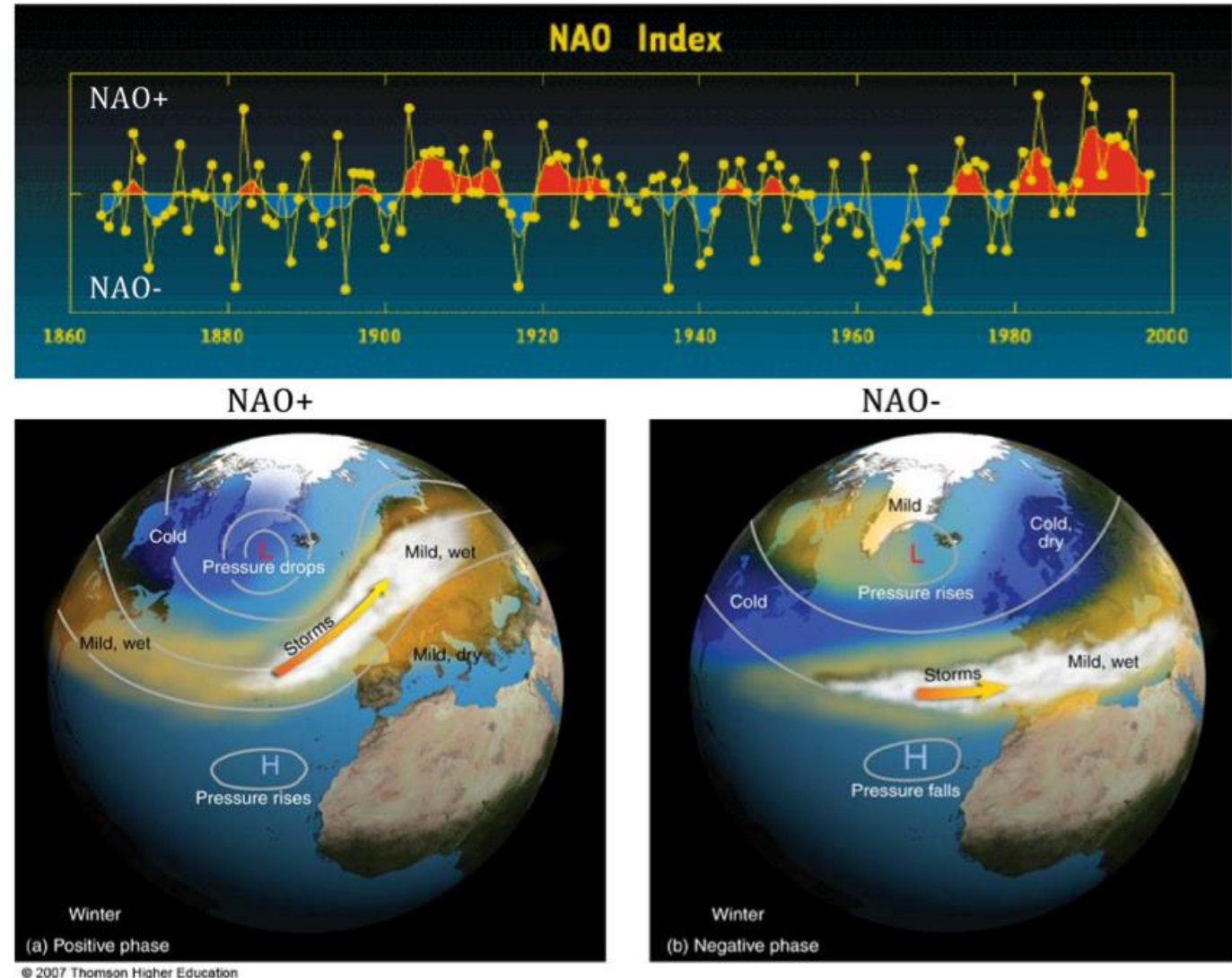
PRACTICE

NAO: North Atlantic Oscillation

NAO index is **based on the surface sea-level pressure** difference between the Subtropical (Azores) High and the Subpolar Low.

The **positive** phase of the NAO reflects below-normal heights and pressure across the high latitudes of the North Atlantic and above-normal heights and pressure over the central North Atlantic, the eastern United States and western Europe.

The **negative** phase reflects an opposite pattern of height and pressure anomalies over these regions. Both phases of the NAO are associated with basin-wide changes in the intensity and location of the North Atlantic jet stream and storm track, and in large-scale modulations of the normal patterns of zonal and meridional heat and moisture transport, which in turn results in changes in temperature and precipitation patterns often extending from eastern North America to western and central Europe.



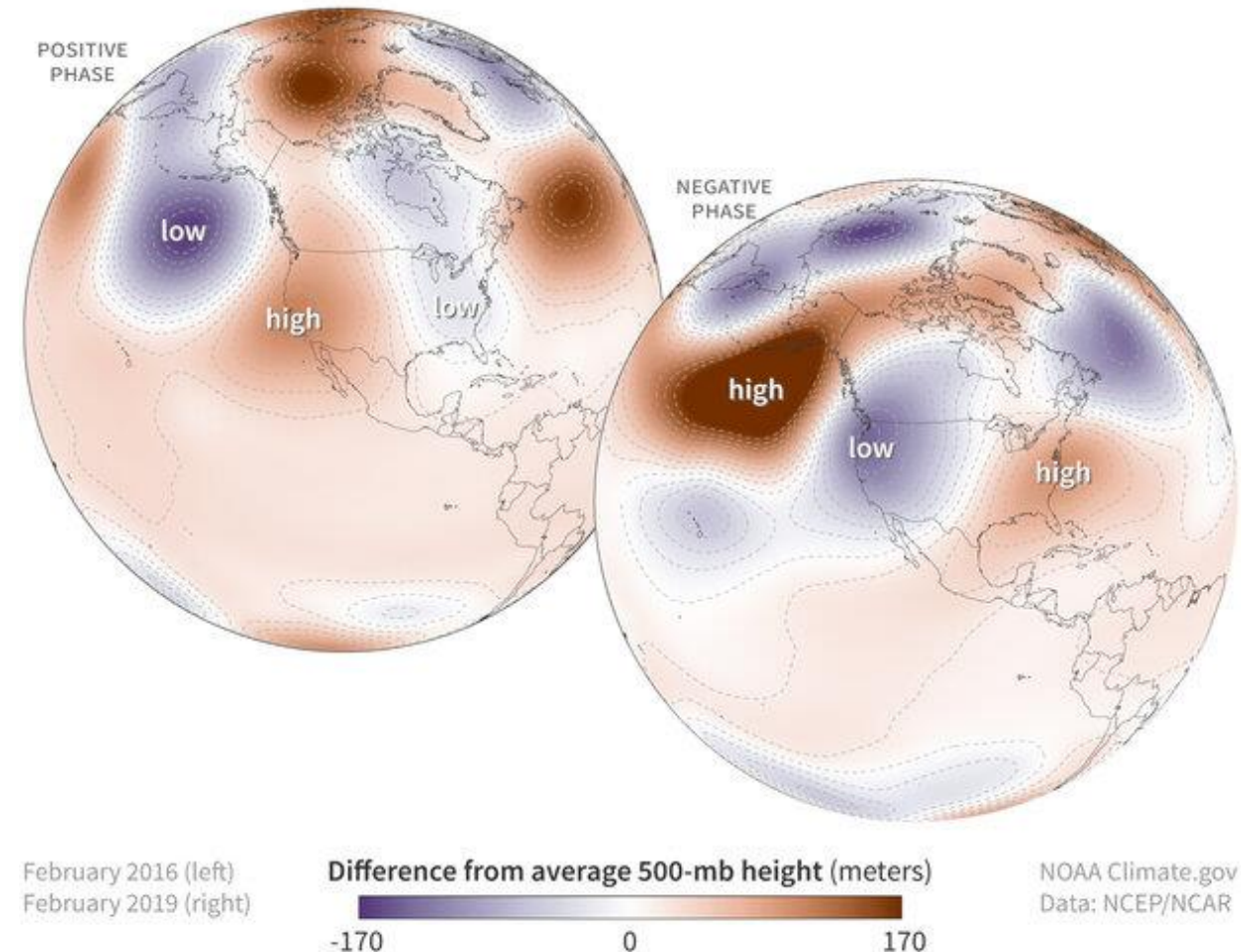
PNA: Pacific-North American

A pattern of air pressure anomalies—departures from the long-term average—at four locations over the Pacific Ocean and North America correlate with regional temperature and precipitation anomalies across North America.

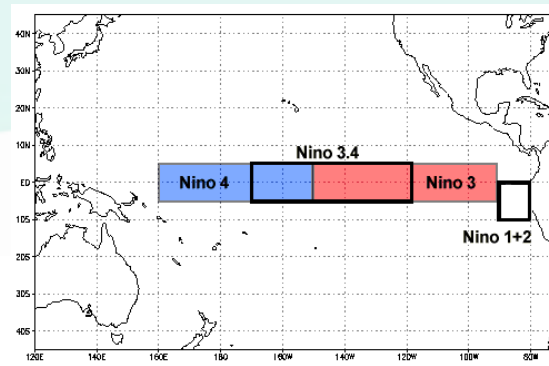
This pattern, known as the PNA, influences regional weather by affecting the strength and location of the East Asian jet stream, and subsequently, the weather it delivers to North America.

The **negative** phase of the PNA pattern is associated with a weaker jet stream across the central Pacific Ocean, high-pressure "blocking" of atmospheric flow in the high latitudes of the North Pacific, and a split-flow of the jet stream over the central North Pacific. Temperature and precipitation departures from normal during negative PNA phases are generally opposite those of the positive phase.

PNA TRI-POLE PRESSURE PATTERNS



N34: Niño 3.4

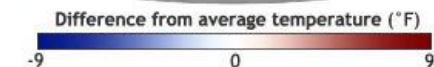
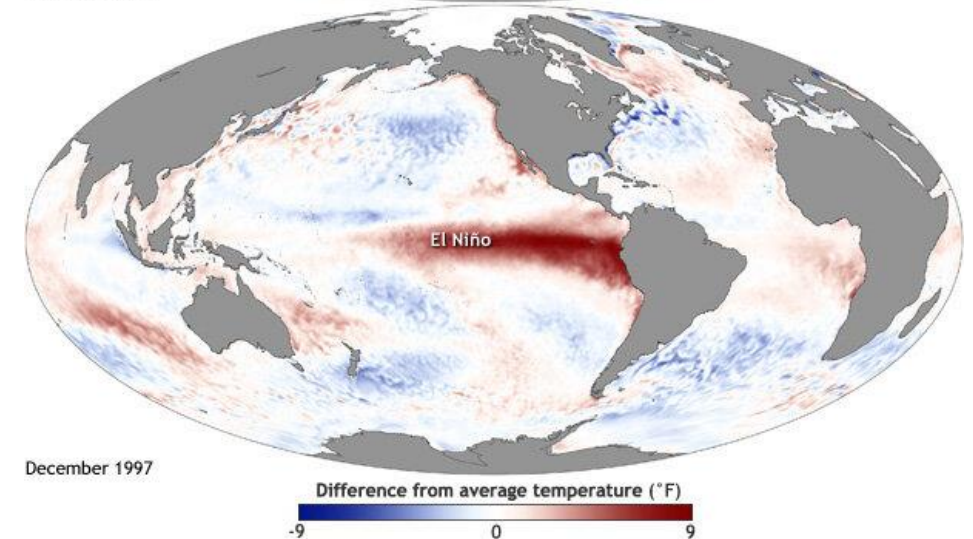
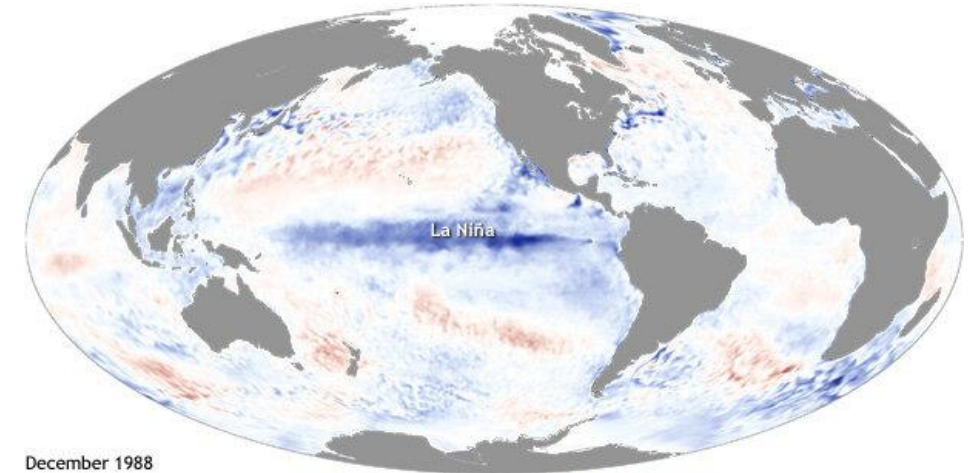


The N34 index and the Oceanic Niño Index (ONI) are the most commonly used indices to define El Niño and La Niña events.

N34 anomalies may be thought of as representing the average equatorial **SSTs** across the Pacific from about the dateline to the South American coast.

The Niño 3.4 index typically uses a 5-month running mean, and El Niño or La Niña events are defined when the Niño 3.4 SSTs exceed $\pm 0.4^{\circ}\text{C}$ for a period of six months or more.

This index was adopted as the primary index for monitoring ENSO due to its ability to capture basin-scale variability



Thank you

jared.ortizangulo@unican.es



<https://geocean.sci.unican.es/hyweb/>



GeoOcean



@GeoOcean_UC

ProCoast 2025
October 20th-23th, 2025

